

CSST 101 - Assessment Task 4

Lecture Notes - Reflection and Discussion

1. Non-monotonic reasoning is closer to human thinking because as people with changing minds, we often change our conclusions when introduced with a new information. We don't instantly know everything about something so we draw assumptions and conclusions based on what we currently know and is possible for changes when new information is discovered that may invalidate our initial assumption and conclusion. Thus, updating our knowledge regarding a certain situation or someone. Non-monotonic reasoning follows the same process, allowing flexible and adaptive reasoning, akin to human thinking.
2. For instance, I was going to heat up some food in a microwave but the light inside wasn't turning on. It was plugged in so I'm confused why it was not working. I concluded that it was broken and told my mother so it could get fixed. My mother told me that there was no electricity in our house.
3. Argumentation frameworks provide a clear line of reasoning through a visual diagram. It shows how arguments oppose or support the conclusion and which arguments conflict with each other. Through this, it is shown which argument is better and its reasoning.

Part 1. Conceptual Understanding

1. Non monotonic reasoning invalidates the previous conclusion when new knowledge is added. This means the outcome isn't fixed and can be changed by a new factor.
2. In monotonic reasoning, the initial conclusion is still true no matter what information is added. While in nonmonotonic, the initial conclusion will be inaccurate and brings a new conclusion when a new information is recognized.
3. An instance would be when you plan to go out with friends and you're in a dilemma of whether bringing an umbrella or just a hat. You checked the sky, it's clear and sunny. An umbrella would be heavy in a bag and a hat would be perfect to wear. You checked the news and discovered that there would be a downspour later in the afternoon. So, you brought an umbrella. In this situation, the initial conclusion was wearing a hat instead because of the clear weather and then when the weather report was known, the decision changed to bringing an umbrella.
4. A default rule is typical conclusion most assume when no further information is known and can be disregarded when a new information is learned. For example, it smells fried chicken in your house so you assume your mother is cooking chicken for lunch. You got excited and went out to see. To your disappointment, your mother wasn't cooking. It was the neighbor's cooking. The default rule is that if it smells fried chicken in the house, then your mother must be cooking. When new information was learned, the default rule is no longer valid and brought out a new conclusion.
5. It helps AI systems decide between conflicting rules by comparing which argument is stronger or more credible. When two rules give different conclusions, the system checks

the reasons and compare them. It then accepts the argument that has better evidence and rejects the weaker one. It helps AI make fair and logical decisions even when rules disagree.

Part 3. Reflection and Discussion

A situation I can recall is when I discovered the Mi Goreng brand. I insisted on coming with my mom on her grocery shopping because I wanted to buy instant Korean ramyeon. When we arrived at the supermarket, I went to the noodles section and grabbed the brand that I wanted. It's cheaper to buy in supermarkets rather than convenience stores. Then, I came across the Mi Goreng brand, and I saw a chicken-flavored one with fried garlic crispies inside. A pack of 5 was much cheaper than the noodles I initially wanted. I decided to buy Mi Goreng instead and I did not regret choosing it because it was delicious and I like the crispies with it. This situation is similar to non-monotonic reasoning in AI, where a system changes its conclusion when new information is introduced. At first, the AI might assume something based on general rules, but it revises its conclusion when it learns new facts that contradict the earlier assumption. Just like I changed my choice after discovering a cheaper option, AI systems also adapt and update their reasoning to make more accurate decisions.